RelCompat3D: Re-Ranking 3D Scene Graph Relations with Geometric Evidence

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Abstract

3D scene graph predictors can rank semantically plausible relations that contradict the reconstructed geometry of the corresponding ordered subject–object pair, such as placing a desk above a ceiling. We introduce RelCompat3D, a post-hoc re-ranking framework that estimates whether a predicted predicate is geometrically compatible with its ordered pair. RelCompat3D estimates compatibility from predicate semantics and predicate-independent geometric measurements while excluding the source relation score and predictor identity. We train lightweight linear and nonlinear estimators using ground-truth relations and constructed counterfactuals. Transformation averaging assigns the same compatibility to equivalent endpoint and predicate representations. At inference time, RelCompat3D combines compatibility with the source relation score to re-rank proximity and vertical-order candidates within their respective families, while support/contact candidates retain their source order. We evaluate fixed predictions from VL-SAT, Open3DSG, and SceneGraphFusion on shared 3DSSG validation scenes using exact-match Recall@K together with verifier-derived Violation@K. Across all reported predictor–K settings, both RelCompat3D variants improve or preserve Recall point estimates while reducing or matching Violation point estimates, with the largest gains observed for Open3DSG. Counterfactual controls show the importance of associating each predicate with the correct ordered pair and its measured geometry. A point- and mesh-based audit further supports the direction of the Violation changes under alternative geometric measurements. These results support predicate–geometry compatibility as a post-hoc reliability signal for fixed 3D scene graph relations.

1 Introduction

3D scene graphs represent reconstructed scenes as structured objects and relations for spatial reasoning and embodied perception (Armeni et al. 2019; Wald et al. 2020). They also support scene alignment (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024) and visual grounding (Liu et al. 2026). A predicted relation must describe the corresponding ordered subject–object pair in 3D, not merely appear plausible at the category or language level.

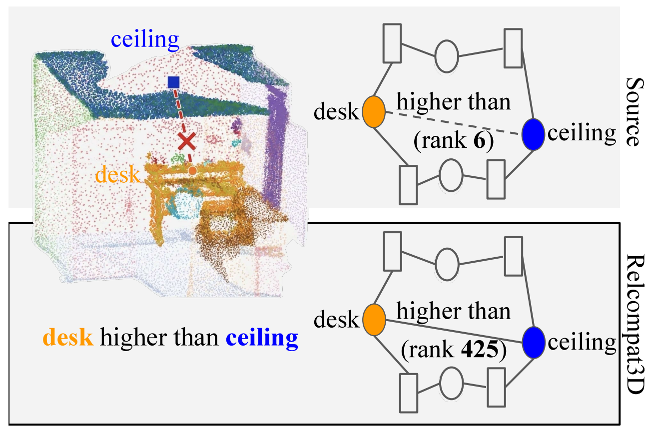

Figure 1: Preview of RelCompat3D. A highly ranked 3D scene graph relation can conflict with the reconstructed geometry of its ordered subject–object pair. Open3DSG (source) (Koch et al. 2024b) ranks "desk higher than ceiling" at 6, although the point cloud places the desk below the ceiling. RelCompat3D-Linear moves it to 425, outside the top-50.

We study a failure mode in which relation predictors assign high scores to predicates that are inconsistent with the measured geometry of the ordered pair. A relation can be plausible for an object-category pair while contradicting measured distance, projected overlap, or vertical arrangement. This mismatch becomes consequential as 3D scene graphs become interfaces for open-vocabulary applications and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Fei et al. 2026; Madhavaram et al. 2026). Figure 1 shows this failure.

Existing predictors already use geometric, edge, and point-cloud evidence (Wald et al. 2020; Zhang et al. 2021). Other methods add multimodal, self-supervised, or open-vocabulary cues (Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). Geometry may inform a predictor without making its source relation score an explicit estimate of whether the same ordered pair satisfies the predicate in 3D. This motivates separating predicate semantics T, ordered-pair measurements G, and the source relation score Z.

We propose RelCompat3D, which preserves the ordered-pair identity of each candidate and learns predicate–geometry compatibility from T and G without using Z or predictor identity. We formulate this separation with family-specific linear heads and a compact shared nonlinear estimator. Both estimators are trained with linked positive–counterfactual pairs so that geometry-consistent candidates outrank their incompatible counterparts. Counterfactuals create incompatible training pairs, whereas the transformations encode alternative representations of the same relation semantics. We then average compatibility over known relation-preserving transformations, including endpoint symmetry for proximity and joint endpoint swap with inverse-predicate transformation for vertical-order. These choices address distinct failure modes. Excluding Z prevents direct copying of the source relation score. Preserving ordered-pair identity ensures that compatibility is computed from the geometry of the corresponding pair. Transformation averaging assigns equal compatibility to equivalent endpoint/predicate representations.

The source relation score is used only during re-ranking. Within proximity and vertical-order relations, RelCompat3D combines it with compatibility and changes the order within each relation family. It preserves the sequence of relation-family labels in the source ranking and the source order of support/contact candidates. Matched fusion baselines and controls examine the roles of the source relation score, ordered-pair identity, and transformation consistency.

We evaluate fixed predictions from Open3DSG (Koch et al. 2024b), VL-SAT (Wang et al. 2023), and a released SceneGraphFusion benchmark model (SGFN) (Wu et al. 2021) on the official 3DSSG validation split. These predictors provide complementary open-vocabulary and closed-set relation predictions. We report exact-match Recall@K and verifier-derived Violation@K on the shared 3DSSG validation scenes. Across all reported K values, both variants match or improve source-ranking Recall point estimates and match or reduce source-ranking Violation point estimates. A separate point- and mesh-based audit evaluates the same rankings using alternative geometric measurements.

In summary, our contributions are threefold:

1. We define and measure a mismatch between source relation scores and ordered-pair geometric compatibility using exact-match Recall@K and verifier-derived Violation@K.
2. We estimate predicate–geometry compatibility without the source relation score and enforce exact consistency under applicable endpoint and predicate transformations.
3. We introduce family-aware re-ranking and characterize its predictor- and relation-family-dependent behavior across three fixed predictors.

2 Related Work

2.1 3D Scene Graph Prediction

3D scene graphs organize reconstructed environments as object nodes connected by semantic and spatial relation edges (Armeni et al. 2019). 3RScan provides repeated indoor reconstructions, and 3DSSG adds object-relation annotations and a relation-prediction benchmark (Wald et al. 2019, 2020). Incremental and online systems study persistent graph construction from RGB-D streams and spatial perception pipelines (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Edge-oriented and multimodal methods improve relation features and spatial reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023). Recent methods study self-supervised, object-centric, and edge-centric representations (Koch et al. 2024a; Heo et al. 2026; Ma et al. 2026). These methods and RelCompat3D both use reconstructed scene evidence, but they optimize relation generation, whereas RelCompat3D re-ranks fixed relation candidates after prediction.

Open-vocabulary 3D perception provides queryable point and instance features (Peng et al. 2023; Takmaz et al. 2023). Graph systems use open-vocabulary features for querying, planning, navigation, and task-driven interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph methods extend this direction to open-vocabulary objects and relations, online generation, and functional reasoning (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). RelCompat3D asks whether an already predicted predicate is supported by the reconstructed geometry of its ordered pair.

2.2 Geometry-aware Relation Evidence

3D relation predictors use edge features, metric geometry, and semantic–geometric cues (Zhang et al. 2021; Feng et al. 2023). Scene graph alignment methods use semantic and geometric structure for matching and registration (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024). Adjacent work uses geometry to constrain 3D object placement or ground object affordances (Huang et al. 2025; Shao et al. 2025). A recent preprint proposes visual–geometric relation witnesses for open-vocabulary relation learning under incomplete supervision (Nguyen et al. 2026). These approaches and RelCompat3D share the goal of grounding semantic relations in geometric evidence. RelCompat3D differs by excluding predictor identity and the source relation score from compatibility and applying the resulting signal to fixed candidates within each relation family.

Several systems incorporate geometry or transformation structure directly into relation generation. GEODE jointly denoises object geometry and relation predicates inside an open-vocabulary graph generator (Feng et al. 2026). RelGraphOV uses relation-aware graph construction and geometric pruning to improve open-vocabulary 3D scene representations (Chen et al. 2026). Transformation-Aware Decoupling modifies the relation generator to improve viewpoint robustness (Sun et al. 2026). For fixed predictions from a 2D scene graph generator, Neau et al. mine spatial, functional, and relational constraints and report a Constraint Violation Rate after declarative refinement (Neau et al. 2026). These methods and RelCompat3D all use structured relation evidence. However, RelCompat3D estimates a continuous compatibility score from the reconstructed ordered pair and enforces applicable endpoint/predicate transformations before joint Recall@K–Violation@K evaluation.

2.3 Reliability Evaluation and Calibration

Confidence calibration asks whether model scores reflect empirical accuracy. Neural network confidence scores often diverge from empirical accuracy, and temperature scaling corrects this divergence without changing the predicted class order (Guo et al. 2017). Deep ensembles estimate predictive uncertainty (Lakshminarayanan, Pritzel, and Blundell 2017). Evaluations under distribution shift test whether that uncertainty remains reliable (Ovadia et al. 2019). Scene-graph work models semantic ambiguity (Yang et al. 2021), calibrates long-tailed predictions (Zhu et al. 2024), and constructs conformal prediction sets (Nag et al. 2025). In 3D, SCR-SSG refines relation scores from masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). PUF propagates uncertainty during online fusion (Yang et al. 2026). These methods and RelCompat3D both address reliability beyond task accuracy, but RelCompat3D measures predicate–geometry compatibility rather than calibrated correctness, predictive uncertainty, or selective abstention.

Recall@K measures exact-match retrieval at a rank cutoff rather than compatibility with reconstructed ordered-pair geometry. RelCompat3D therefore reports exact-match Recall@K with verifier-derived Violation@K and uses compatibility for within-family re-ranking. This joint evaluation complements conventional retrieval and calibration analyses. The two metrics are not calibrated probabilities.

3 Method

RelCompat3D re-ranks fixed relation predictions rather than replacing a 3D scene graph generator. It preserves the ordered-pair identity of each candidate, estimates predicate–geometry compatibility without the source relation score, and combines the two only during re-ranking. We first define the compatibility inputs, then describe the two estimators and the family-aware ranking rule. Figure 2 presents an overview of RelCompat3D.

3.1 Problem Formulation

Let a relation predictor produce a set of relation candidates for a 3D scene. Each candidate is written as $r_i = (\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i, p_i, a_i, Z_i)$. s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate label, a_i is the mapped relation family, and Z_i is the relation score produced by the predictor. We use sigmoid relation scores for VL-SAT (Wang et al. 2023) and SGFN (Wu et al. 2021). For Open3DSG (Koch et al. 2024b), we use cosine similarity between normalized text embeddings. We rank candidates separately for each predictor without predictor-specific normalization or refitting, so existing scores are never compared across predictors.

A context is a scene subgraph over which top-K selection is performed independently. We distinguish the ordered-pair identity $k_i^{\mathrm{pair}} = (\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i)$ from the exact relation-candidate identity $k_i^{\mathrm{rel}} = (\mathrm{scan}_i, \mathrm{context}_i, s_i, p_i, o_i)$ used for evaluation. The ordered pair is retained when associating each prediction with geometry and ground truth.

We represent predicate semantics as $T_i = p_i$. The family label a_i selects the applicable transformations and determines whether the candidate is re-ranked. For the Linear estimator, it also selects the compatibility head and training-split normalization statistics. The shared nonlinear estimator instead encodes the family as an input indicator. We then associate predicate-independent geometric measurements of the ordered pair $G_i = G(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i)$.

The reported relation families are $\mathcal{A}_{\mathrm{eval}} = \{\mathrm{support/contact},\ \mathrm{proximity},\ \mathrm{vertical\,order}\}$, whose consistency can be evaluated from reconstructed ordered-pair geometry. Candidates outside these families are excluded from the reported Recall and Violation evaluation. The re-ranking relation families are narrower: $\mathcal{A}_{\mathrm{rank}} = \{\mathrm{proximity}, \mathrm{vertical\,order}\}$, for which the defined endpoint/predicate transformations are available. Support/contact is evaluated but kept in source order because this family requires richer local contact and pose evidence. No single endpoint transformation preserves the semantics for every predicate in the family.

For estimator $q \in \{\mathrm{lin}, \mathrm{mlp}\}$, our objective is to estimate a bounded compatibility score

$$C_i^q = \sigma(f_q(T_i, a_i, G_i)) \in [0, 1]. \tag{1}$$

Here σ denotes the logistic sigmoid. We then average over the valid family-specific endpoint/predicate transformations to obtain the transformation-consistent score $C_i^{\mathrm{tr},q}$. The source relation score is introduced only in the within-family ranking score defined below. C_i^q is a compatibility score used for ranking, not a posterior probability of geometric validity. Training-split ground truth provides positives. Proximity negatives use distant, non-overlapping same-context pairs. Vertical-order negatives keep the endpoints fixed and replace the predicate with its inverse. Separately, relation-preserving augmentation swaps proximity endpoints without changing the predicate and jointly swaps vertical endpoints with the inverse predicate. Evaluation rows, source relation scores, and verifier-status labels are not used to construct training examples.

3.2 Compatibility Estimation

For each candidate relation r_i, the associated geometry vector G_i contains oriented bounding box (OBB)-derived 3D/XY distances, relative height, projected overlap, and vertical-gap features. Point-level contact evidence is used only by the support/contact verifier, not by the compatibility model.

Let $\mathrm{std}_{\mathrm{tr}}$ standardize each numeric feature with training-split statistics and replace a missing value by the corresponding training mean. Both estimators preserve the same separation between compatibility inputs and the source relation score and use the same training targets, loss, and transformation averaging.

Linear Estimator. The feature map for RelCompat3D-Linear is

$$\begin{aligned} &\Phi_{\mathrm{lin}}(T_i, G_i) = [1; \phi_T(T_i); \mathrm{std}_{\mathrm{tr}}(G_i); \mathrm{std}_{\mathrm{tr}}(\phi_\times(T_i, G_i))], \\ &\phi_\times(T_i, G_i) = [d(p_i)\Delta z_i,\ d(p_i)\Delta z_i^{\mathrm{norm}}]. \end{aligned} \tag{2}$$

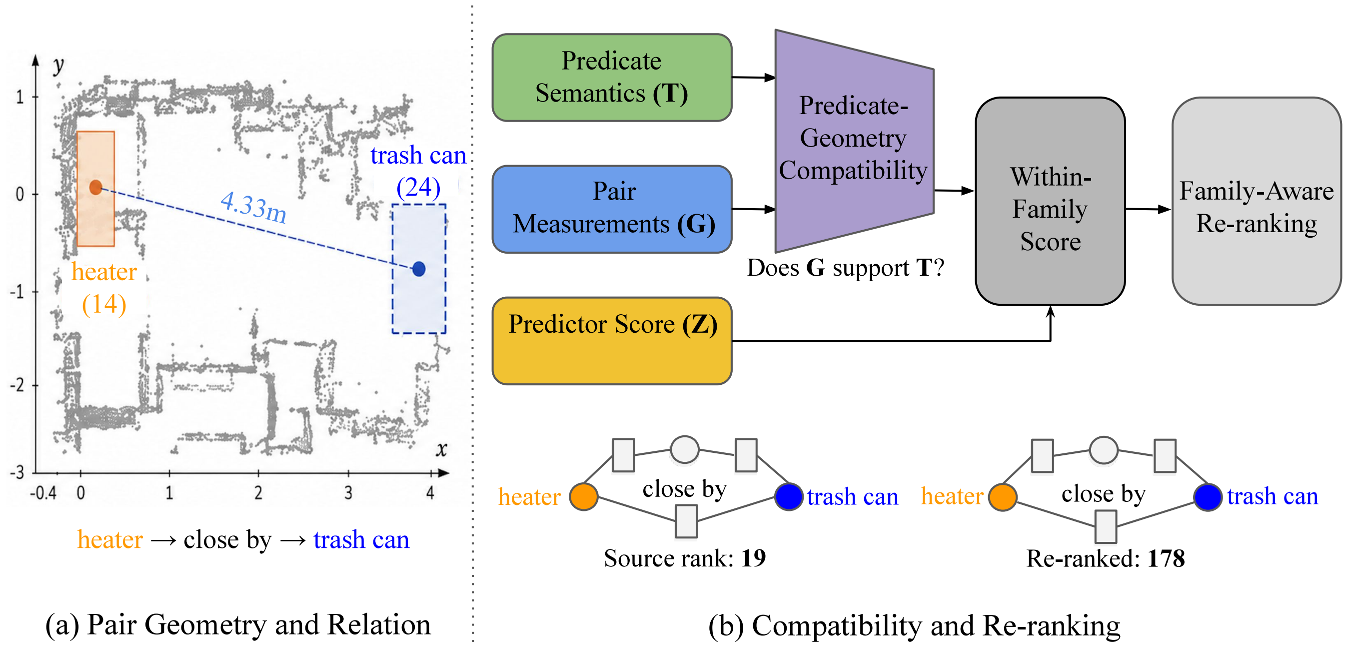

Figure 2: RelCompat3D overview. (a) Open3DSG ranks "heater close by trash can" at 19, although the reconstructed ordered pair has an XY center distance of $4.33\,\mathrm{m}$. (b) RelCompat3D estimates transformation-consistent predicate–geometry compatibility from predicate semantics and ordered-pair measurements, without using the source relation score. The source relation score is combined with compatibility only during within-family scoring, after which RelCompat3D-Linear moves the relation from rank 19 to 178.

Let z_{s_i} and z_{o_i} denote the vertical coordinates of the subject and object OBB centers, and let h_{s_i} and h_{o_i} denote their OBB heights. We define $\Delta z_i = z_{s_i} - z_{o_i}$ and $\Delta z_i^{\mathrm{norm}} = 2\Delta z_i/(h_{s_i} + h_{o_i})$. Here ϕ_T contains predicate indicators, and $d(p_i)$ is 1 for `higher than`, -1 for `lower than`, and 0 otherwise. The family label a_i selects the linear head and training-split normalization statistics but is not included as a constant input: $f_{\mathrm{lin}}(T_i, a_i, G_i) = w_{a_i}^\top \Phi_{\mathrm{lin}}(T_i, G_i)$.

Nonlinear Estimator. RelCompat3D-MLP uses one shared multilayer perceptron (MLP) with a single two-unit ReLU hidden layer. Its input Ψ_i contains family and predicate indicators, the same underlying geometric measurements, the two signed-height interactions above, and the sum of the two directional overlap ratios. With a predicate-linear skip connection, its logit is

$$\begin{aligned} h_i &= \mathrm{ReLU}(W\Psi_i + b), \\ f_{\mathrm{mlp}}(T_i, a_i, G_i) &= v^\top h_i + b_o + \beta^\top \phi_T(T_i). \end{aligned} \tag{3}$$

Here W and v are learned weights, b and b_o are biases, and β weights the predicate-indicator skip path. The skip connection provides a direct linear path from predicate indicators to the output logit. The nonlinear estimator is shared across families, so its family indicator is not a constant feature. Both estimators exclude the source relation score, rank, predictor identity, and object-class features.

For proximity and vertical order, H_a contains the identity and the family-specific transformation τ_a, defined as follows. For proximity, τ_a swaps the ordered endpoints and leaves `close by` unchanged. For vertical order, τ_a swaps the endpoints and maps `higher than` ↔ `lower than`. Transformation averaging therefore yields $C^{\mathrm{tr,q}}_{\mathrm{close}}(s,o) = C^{\mathrm{tr,q}}_{\mathrm{close}}(o,s)$ and $C^{\mathrm{tr,q}}_{\mathrm{higher}}(s,o) = C^{\mathrm{tr,q}}_{\mathrm{lower}}(o,s)$. For support/contact $H_a = \{e\}$, where e is the identity transformation. Its compatibility head is used only for the Product (all families) comparison. Training combines this augmentation with a linked positive–counterfactual ranking loss. For every linked positive–counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logits $\ell_i^q = f_q(T_i, a_i, G_i)$ receive a margin penalty:

$$\begin{aligned} \mathcal{L}_q = \mathcal{L}_{\mathrm{BCE}} &+ \lambda_{\mathrm{pair}}\, \mathbb{E}_{\mathcal{P}} \left[\mathrm{softplus}\left(m - (\ell^q_{i+} - \ell^q_{i-}) \right) \right] \\ &+ \lambda_{\mathrm{reg}} \mathcal{R}(\theta_q). \end{aligned} \tag{4}$$

Here $\mathcal{L}_{\mathrm{BCE}}$ is binary cross-entropy over positive and counterfactual examples. The second term is a softplus margin-ranking loss with margin m. We set $m = 1$, $\lambda_{\mathrm{pair}} = 0.25$, and $\lambda_{\mathrm{reg}} = 10^{-4}$ in all experiments. These values are shared across estimators and predictors without predictor-specific search. $\mathcal{R}$ is an ℓ_2 penalty on the non-bias parameters in θ_q, where θ_q denotes the trainable parameters of estimator q. Both estimators are fitted exclusively on the training split.

Finally, let H_{a_i} denote the finite transformation group for family a_i, and define the transformation orbit $\mathcal{O}_i = \{g(T_i, G_i) : g \in H_{a_i}\}$. We average the learned score over this set to obtain the transformation-consistent compatibility:

$$C_i^{\mathrm{tr,q}} = \frac{1}{|\mathcal{O}_i|} \sum_{(T',G') \in \mathcal{O}_i} \sigma(f_q(T', a_i, G')). \tag{5}$$

Group averaging makes proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation exact at inference. The pairwise objective encourages each linked positive to score above its counterfactual.

3.3 Family-Aware Re-Ranking

For each compatibility estimator q, define the within-family ranking score

$$u_i^q = \begin{cases} Z_i C_i^{\mathrm{tr,q}}, & a_i \in \mathcal{A}_{\mathrm{rank}}, \\ Z_i, & a_i = \mathrm{support/contact}. \end{cases} \tag{6}$$

Let $\pi^Z = (e_1, \ldots, e_n)$ denote the source ranking, and let a_j be the relation family at position j. For each $a \in \mathcal{A}_{\mathrm{rank}}$, let π_a^q contain the candidates in family a sorted by u_i^q, with ties resolved by exact relation-candidate identity. For support/contact, π_a^q is the corresponding family subsequence of π^Z. Thus, a candidate can exchange positions only with candidates from the same re-ranked family. At position j, the rule selects the next unused candidate from $\pi_{a_j}^q$. The output therefore preserves the source family-label sequence and the support/contact subsequence at every K.

RelCompat3D-Linear and RelCompat3D-MLP differ in how they model compatibility. The first uses family-specific linear heads, and the second uses a shared nonlinear estimator. Rank-fusion baselines use the same ranking procedure with alternative score combinations. A support/contact head is fitted only for the Product (all families) comparison. The supplement provides the complete counterfactual rules, optimization details, and proofs.

4 Experiments

4.1 Experimental Setup

Datasets and Evaluation. Predictions come from Open3DSG (Koch et al. 2024b), VL-SAT (Wang et al. 2023), and SGFN (Wu et al. 2021). We evaluate all three on the 3DSSG validation split of 3RScan (Wald et al. 2019, 2020), restricted to relation families whose consistency can be assessed from reconstructed ordered-pair geometry. All evaluations use the same scope: 157 scans, 548 relation contexts, and 3,972 exact-match ground-truth relations from support/contact, proximity, and vertical-order families. Evaluation-split ground-truth relations are used only when computing Recall@K, never to filter or rank candidate relations.

Baselines and Training. We evaluate the two compatibility estimators defined in Section 3.2 with identical training rows, constructed targets, split, objective, and family-aware ranking procedure. Their fitted parameters are applied to all three predictors without predictor-specific refitting. Matched rank-average fusion (RankAvg) averages the predictor and linear-compatibility ranks. Reciprocal rank fusion (RRF) (Cormack, Clarke, and Buettcher 2009) combines the same rankings using reciprocal ranks. Product (all families) additionally applies the linear support/contact head. Controls perturb the predicate, pair identity, geometric measurements of the ordered pair, endpoint order, source relation score, or distance measurements. We apply the principal controls to both proposed estimators. Normalization, imputation, and model parameters use only 1,061 training scans. Model design and the applicable relation families were selected on the 117-scan development split. The 157-scan evaluation split is never used for fitting. All comparisons across predictors use this shared target.

Metrics. Let $L_K(c)$ be the selected top-K candidates in context c and Y_c its ground-truth relations. Both sets use exact relation identity (s_i, p_i, o_i) within c. Family mapping is used only for re-ranking and never for label matching. We report

$$\mathrm{Recall@K} = \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}. \tag{7}$$

Let $D_K \subseteq \bigcup_c L_K(c)$ contain selected candidates for which the rule-based geometry verifier returns satisfied, uncertain, or violated, and let N_s, N_u, N_v denote their respective counts. The primary verifier-derived metric is

$$\mathrm{Violation@K} = \frac{N_v}{N_s + N_u + N_v}. \tag{8}$$

Thus uncertain candidates enter the denominator but not the numerator. The supplement reports uncertainty, coverage, decidable-only Violation, and a variant that counts uncertain candidates as violations. We report Recall@K and Violation@K metrics over $K \in \{5, 10, 20, 50, 100\}$. Paired 95% confidence intervals use 1,000 bootstrap resamples of the 157 scans, keeping all contexts from each scan together and using the same resamples across conditions. The supplement reports per-family metrics and family composition at $K = 100$.

4.2 Results

Recall–Violation Results. Table 1 and Figure 3 show that, across $K \in \{5, 10, 20, 50, 100\}$, both RelCompat3D variants have Recall point estimates no lower and Violation point estimates no higher than Source for all three predictors. At $K = 50$, the paired 95% bootstrap intervals are positive for Recall and negative for Violation on Open3DSG and SGFN. On VL-SAT, whose Source Recall is already high, the Recall intervals contain zero while the Violation intervals remain negative, supporting lower verifier-derived Violation without a detectable Recall loss. SGFN is tied with Source at $K = 5$ and changes only marginally at $K = 10$. Open3DSG shows the largest changes, indicating that its Source ranking places more verifier-violated candidates near the top.

No single ranking rule dominates across all predictors and reported K values. At $K = 50$, RelCompat3D-MLP achieves higher Open3DSG Recall than RelCompat3D-Linear (46.70% versus 44.18%) but also higher Violation (4.13% versus 3.42%). The variants are nearly tied on VL-SAT and SGFN. RankAvg and RRF attain lower Violation in some settings but show larger low-K Recall losses on VL-SAT and SGFN. Product (all families) can improve the aggregate metrics but changes support/contact selections.

The Source-relative conclusion remains unchanged for RelCompat3D-Linear under all five pre-specified monotonic score transformations and for RelCompat3D-MLP in all but one comparison. At $K = 50$, both variants also achieve

Predictor	Ranking rule	$K=5$ R↑	$K=5$ V↓	$K=10$ R↑	$K=10$ V↓	$K=20$ R↑	$K=20$ V↓	$K=50$ R↑	$K=50$ V↓	$K=100$ R↑	$K=100$ V↓
Open3DSG	Source	3.42	52.05	9.87	32.89	19.89	20.99	40.43	13.87	51.11	12.42
	RelCompat3D-Linear	3.73	**0.94**	11.38	**2.33**	23.62	**3.13**	44.18	**3.42**	56.85	**3.24**
	RelCompat3D-MLP	3.70	4.95	**11.78**	4.97	**24.67**	4.56	**46.70**	4.13	**59.89**	3.71
	RankAvg	**3.78**	1.13	11.05	2.51	22.36	3.55	43.20	4.60	54.00	5.49
	RRF	3.58	19.59	10.67	16.47	21.58	13.60	42.45	9.60	54.68	7.80
	Product (all families)	9.94	3.98	19.64	4.54	32.80	4.37	46.42	2.69	60.05	3.31
VL-SAT	Source	41.94	0.29	63.22	0.82	80.74	1.42	92.72	2.68	96.35	4.76
	RelCompat3D-Linear	42.07	**0.15**	63.39	0.57	80.82	1.14	**92.77**	1.97	96.58	2.95
	RelCompat3D-MLP	**42.09**	**0.15**	**63.47**	0.51	**80.92**	1.09	92.72	1.89	96.50	2.96
	RankAvg	35.32	**0.15**	51.74	**0.47**	69.41	**0.99**	90.23	**1.62**	**97.05**	**2.26**
	RRF	36.81	**0.15**	55.66	0.51	75.08	1.07	91.72	1.82	**97.05**	2.50
	Product (all families)	41.74	0.11	63.32	0.49	80.94	1.10	92.93	2.03	96.88	3.25
SGFN	Source	**31.17**	**2.37**	**39.75**	3.49	49.12	3.22	74.02	3.85	92.35	6.30
	RelCompat3D-Linear	**31.17**	**2.37**	**39.75**	**3.47**	49.14	2.97	74.50	2.63	**93.03**	3.50
	RelCompat3D-MLP	**31.17**	**2.37**	**39.75**	**3.47**	**49.19**	2.96	**74.57**	2.58	92.88	3.50
	RankAvg	31.09	**2.37**	38.80	**3.47**	46.63	**2.95**	70.22	**2.35**	92.72	**2.60**
	RRF	31.14	**2.37**	38.95	**3.47**	47.08	2.97	68.50	2.54	87.99	3.07
	Product (all families)	31.07	2.48	40.56	3.19	51.18	2.76	77.04	2.58	94.13	3.71

Table 1: Shared 3DSSG validation results. Exact-match Recall (R↑) and verifier-derived Violation (V↓) are reported at each K. All values are percentages. Source denotes the original ranking of each predictor. Bold marks the best value among rows that preserve the source family sequence and support/contact order, including ties. Product (all families) re-ranks every relation family and is reported as a scope comparison.

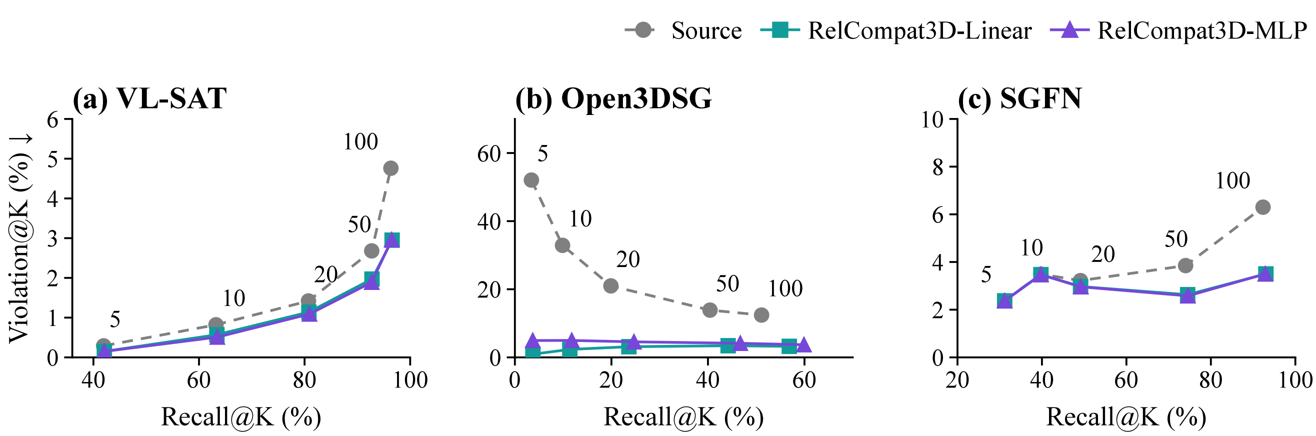

Figure 3: Recall@K–Violation@K trajectories. All three predictors are evaluated on the shared 3DSSG validation scenes. Each line connects $K \in \{5, 10, 20, 50, 100\}$ for Source, RelCompat3D-Linear, and RelCompat3D-MLP. Rightward and downward movement is preferred. Numbers label K along the Source trajectory. All curves follow the same increasing-K order. Axis ranges differ by predictor.

higher Recall and lower Violation than the non-learned robust-density baseline for all three predictors. Matched routing controls produce estimator- and K-dependent results when proximity and vertical-order candidates are re-ranked together. We therefore treat family-aware re-ranking as a composition-preserving constraint rather than an aggregate-optimal rule.

The qualitative results in Figure 1 and Figure 2 show that family-aware re-ranking demotes geometrically inconsistent candidates in both vertical-order and proximity. The supplement provides the complementary case of an exact-match, verifier-satisfied proximity candidate entering the top-50.

Ablations and Controls. Table 2 tests whether the ranking gains depend on the predicate, the corresponding ordered pair, and the geometry measurements. For RelCompat3D-Linear, wrong-pair and shuffled geometry reduce Recall and increase Violation for every predictor at both reported K values. These controls show that the compatibility estimate is

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	**RelCompat3D-Linear**	44.18	3.42	56.85	3.24
	RelCompat3D-MLP	46.70	4.13	59.89	3.71
	Wrong predicate	42.65	21.98	53.47	20.96
	Wrong pair	38.52	8.50	49.32	8.36
	Shuffled geometry	38.42	12.93	48.79	12.42
	Fixed-predicate swap	42.67	21.98	53.63	20.96
	Distance only	51.16	8.24	63.22	9.55
	Compatibility only	42.22	3.42	56.87	3.24
VL-SAT	**RelCompat3D-Linear**	92.77	1.97	96.58	2.95
	RelCompat3D-MLP	92.72	1.89	96.50	2.96
	Wrong predicate	90.46	4.97	94.79	7.95
	Wrong pair	90.56	2.61	94.96	4.73
	Shuffled geometry	89.17	3.05	94.11	5.60
	Fixed-predicate swap	90.46	4.97	94.81	7.95
	Distance only	81.90	5.34	89.80	8.09
	Compatibility only	67.65	1.61	83.89	2.03
SGFN	**RelCompat3D-Linear**	74.50	2.63	93.03	3.50
	RelCompat3D-MLP	74.57	2.58	92.88	3.50
	Wrong predicate	71.58	9.42	89.93	12.99
	Wrong pair	71.55	3.99	88.07	6.65
	Shuffled geometry	70.59	4.73	86.83	8.36
	Fixed-predicate swap	71.58	9.42	90.03	12.98
	Distance only	63.19	10.00	84.06	12.66
	Compatibility only	52.79	2.32	70.64	2.30

Table 2: Ablations and counterfactual controls. All conditions are evaluated on the shared 3DSSG validation scenes. For compactness, control rows are shown for RelCompat3D-Linear. RelCompat3D-MLP rows report the full nonlinear variant for comparison, with matched MLP controls in the supplement. R is Recall (↑) and V is Violation (↓). All entries are percentages. Every condition uses the same candidates and family-aware ranking procedure as Table 1.

sensitive to the association between a candidate and the geometry of its corresponding ordered pair. A wrong predicate and an endpoint swap that keeps the predicate fixed increase Violation. Their nearly identical aggregate values are consistent with both controls reversing the signed interpretation of vertical-order geometry.

For RelCompat3D-Linear, distance-only ordering raises Open3DSG Recall but substantially increases its Violation, while degrading both metrics on VL-SAT and SGFN. Compatibility-only ordering preserves or lowers Violation but loses Recall on VL-SAT and SGFN, showing that compatibility does not replace the source relation score. The supplement reports feature-removal analyses, direct component removals, transformation checks, and matched controls for both estimators.

Point- and Mesh-Based Consistency Audit. The compatibility estimators and primary verifier share some oriented bounding box (OBB)-derived measurements. We therefore remeasure proximity and vertical-order violations directly from reconstructed instance geometry using two representations. One estimate uses robust point-cloud vertex distances and heights and the other uses area-weighted samples from reconstructed mesh triangles. Thresholds are fixed

Predictor	Source	Linear	ΔV	Coverage (M/D)
VL-SAT	16.43	**14.34**	-2.09	95.5 / 71.0
Open3DSG	45.96	**4.90**	-41.06	98.2 / 83.7
SGFN	11.56	**8.13**	-3.43	95.8 / 79.9

Table 3: Point- and mesh-based consistency audit. Results are reported at $K = 50$ on the shared 3DSSG validation scenes. A candidate is labeled satisfied or violated only when the two measurements agree, and disagreement is labeled uncertain. Source and Linear report Violation percentages for the source ranking and RelCompat3D-Linear. ΔV is Linear minus Source in percentage points. Coverage reports measured and decidable percentages (M/D).

from training-split positive relations. The audit labels do not use OBB measurements, predictor identity, source relation score, learned compatibility, or primary verifier status labels.

Table 3 reports the $K = 50$ audit for RelCompat3D-Linear. At $K = 50$, RelCompat3D-Linear lowers the alternative Violation rate for all three predictors with paired intervals below zero. The supplement reports both variants at all five K values. All changes are reductions except the SGFN tie at $K = 5$. The audit changes only the evaluation labels, so Recall remains as reported in Table 1.

5 Discussion and Limitations

All three predictors are evaluated on the same 3DSSG validation split (Wald et al. 2020). The results therefore characterize behavior across predictors on shared data rather than dataset-level generalization. RelCompat3D assumes known instances and reconstructed ordered-pair geometry, and it retains support/contact candidates in source order.

Broader claims require independently defined reference labels, richer contact and pose evidence, and evaluation on additional datasets. Because the point- and mesh-based audit uses the same reconstructed scenes and ontology, it is an alternative geometric measurement rather than independent ground truth for geometric validity.

6 Conclusion

RelCompat3D addresses high-scoring predicates that conflict with reconstructed ordered-pair geometry by learning predicate–geometry compatibility separately from the source relation score and re-ranking candidates within applicable relation families. Across three predictors on the shared 3DSSG validation scenes, both variants show lower or tied verifier-derived Violation point estimates while preserving or improving Recall point estimates at every reported K.

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